Lab Assignment 2

Data Manipulation Using Python and Pandas

Course: Data Engineering DS5003A

Objective

The objective of this laboratory assignment is to familiarize students with reading data from a CSV file, manipulating tabular data using the Pandas library, writing reusable Python functions, applying lambda functions, and solving simple data analysis problems.

Dataset Description

Use the file **students.csv**.

The dataset contains the following columns:

- Roll_no
- Name
- Dept
- Maths
- Physics
- Programming
- Attendance

The dataset consists of 10 student records.

Questions

Q1. Reading the Dataset

Import the Pandas library and load the `students.csv` file into a DataFrame.

Display:

- First 5 rows
- Last 5 rows

Q2. Exploring the Dataset

Display the following information:

- Shape of the DataFrame
- Column names
- Data type of each column

Q3. Selecting and Filtering Data

Perform the following operations:

1. Display only the **Name** and **Programming** columns.
2. Display all students belonging to the **CSE** department.
3. Display students whose attendance is greater than or equal to **75%**.

Q4. Writing Functions

Write the following Python function.

```
def total_marks(maths, physics, programming):
    pass
```

The function should return the total marks of a student.
Create a new column named **Total** using this function.

Q5. Average Marks

Write the following Python function.

```
def average_marks(total):
    pass
```

The function should return the average marks of a student.
Create a new column named **Average** using this function.

Q6. Using Lambda Functions

Using a lambda function, create a new column named **Result**.
Rules:

- Pass if Average ≥ 50
- Fail otherwise

Q7. Grade Calculation

Using a lambda function, create a new column named **Grade** according to the following criteria.

Average Marks	Grade
90–100	A
80–89	B
70–79	C
60–69	D
Below 60	F

Q8. Dictionary

Create a dictionary where

- Key → Roll_no
- Value → Name

Display the dictionary and demonstrate how to retrieve the name of a student using the roll number.

Q9. Department Performance Analysis

Write the following function.

```
def department_report(df, dept):
    pass
```

The function should accept a department name and return:

- Total number of students
- Average Programming marks
- Highest Programming marks

Example:

```
department_report(df, "CSE")
```

Q10. Students Requiring Extra Support

A student requires extra academic support if **either** of the following conditions is true:

- Average marks are below 60.
- Attendance is below 75%.

Display a DataFrame containing only these students.

Question 13

Create a students class with the following information

- Roll_no (given during object creation)
- Name (given during object creation)
- Dept (given during object creation)
- Maths (only be updated via function)
- Physics (only updated via function)
- Programming (only be update via function)
- Attendance (only updated via function)
- a function to check if a student eligible for a scholarship. A student is eligible for a scholarship if
  - Average marks are 85 or above.
  - Attendance is 90% or above.
- a function that checks the performance and issue warning if the marks are lower than 30% in all courses.

create a list of students using the defined data structure from the student.csv file.

Q12. Student Performance Report

Write a function that generates a summary report of the students list.

The report should include:

- Total number of students
- Department-wise student count
- Highest total marks
- Average attendance
- Number of students passed
- Number of students failed
- Name of the top performer

A sample output is shown below:

```
---------- Student Report ----------

Total Students        : 10
Top Performer         : Alice
Highest Total Marks   : 275
Students Passed       : 9
Students Failed       : 1

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```

Question 13 (Class-Based Programming)

Implement a class stack using an array. Add 13, 13, 78, 67 and pop 2 times.

Question 14 (Class-Based Programming)

Implement a stack using a linked-list. Push 13, 13, 78, 67 and pop 2 times.

End of Assignment